

DAVID LÓPEZ DIEGO

Materials & Metrology Scientist | Nanomaterials Characterization & Scientific Data Analysis

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PROFESSIONAL SUMMARY

Physicist and PhD candidate with over four years of experience in advanced materials characterization, nanoscale metrology, scientific instrumentation and experimental data analysis. I work across multidisciplinary fields, including nanomaterials, semiconductors, biosensors, and thermoelectrics, where I bridge physical laboratory operations with computational data workflows. By ensuring data integrity through strict calibration and quality protocols, I provide the reliable foundation needed for R&D projects and the publication of over 10 peer-reviewed papers.

TECHNICAL SKILLS

Materials Characterization: AFM, SEM-EDX, TEM, XRD, Raman, FTIR, XPS, FIB lithography, optical & fluorescence microscopy, HPLC-MS, ICP

Data Analysis & Programming: Python, Scikit-learn, TensorFlow, R, Excel, Origin, ImageJ, Git, Mathematica

Instrumentation & Lab Ops: Calibration, preventive maintenance, data acquisition, cleanroom environment, bacterial cultures, equipment logs

Quality & Documentation: GLP/GMP principles, ISO 9001, workflow standardization, procurement and budget tracking

Additional Tools: SQL, CST Studio Suite, AutoCAD, C++, NI LabVIEW, 3D printing, Power BI

Core competencies: Materials characterization, nanoscale metrology, scientific instrumentation, R&D technical reporting

PROFESSIONAL EXPERIENCE

Materials Scientist - Institute of Micro and Nanotechnology, IMN-CNM, CSIC | Madrid, Spain | Feb 2022 - Present

- Develop AFM methods for nanoscale characterization of nanomaterials, semiconductors, biosensors and biological systems. Establish standardized measurement workflows adopted by 4 research groups, ensuring consistency across multidisciplinary projects.
- Operate, calibrate and maintain 10+ characterization techniques (AFM, SEM-EDS, TEM, FIB, XRD, HPLC-MS, Raman, XPS, ICP, FTIR). Provide instrumental support to 5 research teams and external collaborators, ensuring consistent data quality for ongoing R&D projects.
- Design end-to-end research workflows, from raw data acquisition to processed results and technical reporting, replacing ad-hoc manual processing with standardized and reproducible pipelines.
- Co-author 10+ peer-reviewed publications in materials characterization, nanoscience, biosensors and applied research.
- Actively participate in national and industry-linked R&D initiatives, including PERTE Chip (semiconductor characterization), qNanoFoPho (photonic nano-structures), and biosensor development for early autism detection, providing experimental measurements.
- Perform electrical and thermal transport characterization of thermoelectric materials using Hall, LFA, DSC, TGA, Seebeck, NanoTR and PicoTR measurement systems.
- Maintain ISO 9001-certified laboratory operations through GLP/GMP principles: manage equipment logs, calibration records, SOP documentation and audit preparation, ensuring successful annual quality system reviews.
- Develop Python-based workflows for scientific data analysis, numerical modelling and technical reporting, replacing repetitive manual processing with standardized and reproducible procedures.
- Manage project-related procurement, supplier coordination and budget tracking to support timely delivery of research milestones.
- Train 5+ junior researchers on analytical techniques, sample preparation and instrumentation workflows, enabling them to conduct independent measurements and data analysis within research projects.

Research Assistant, Materials Physics - SMAP Group, Surfaces and Porous Materials | Valladolid, Spain | Jan 2021 - Sep 2021

- Synthesized and characterized 10+ mixed-matrix membrane and polymer formulations for gas-separation applications, optimizing permeability/selectivity/performance trade-offs for CO₂ and greenhouse-gas capture use cases.
- Managed laboratory operations, instrument maintenance and safety protocols in a materials research environment.

EDUCATION

PhD Candidate in Condensed Matter Physics, Nanoscience and Biophysics - Autonomous University of Madrid, Spain | Expected: 2027

MSc in Physics (Materials Physics) - University of Valladolid | 2021

BSc in Physics - University of Valladolid | 2020

CERTIFICATIONS & TRAINING

- Data Science, Machine Learning & GenAI** (Python, R, SQL) – Harvard, Columbia & Udemy (2024-2025)
- Computational Physics** (Mathematica, C++, NI LabVIEW) – University of Valladolid (2021-2022)

LANGUAGES

Spanish: Native | English: C1